Model Results Comparison Table

Comparison	Excel	OpenRocket	Difference	Difference (%)
Apogee (m)	3588.962	3341.552	247.41	7.40%
Max q (kPa)	90.567	79.99	10.577	13.22%

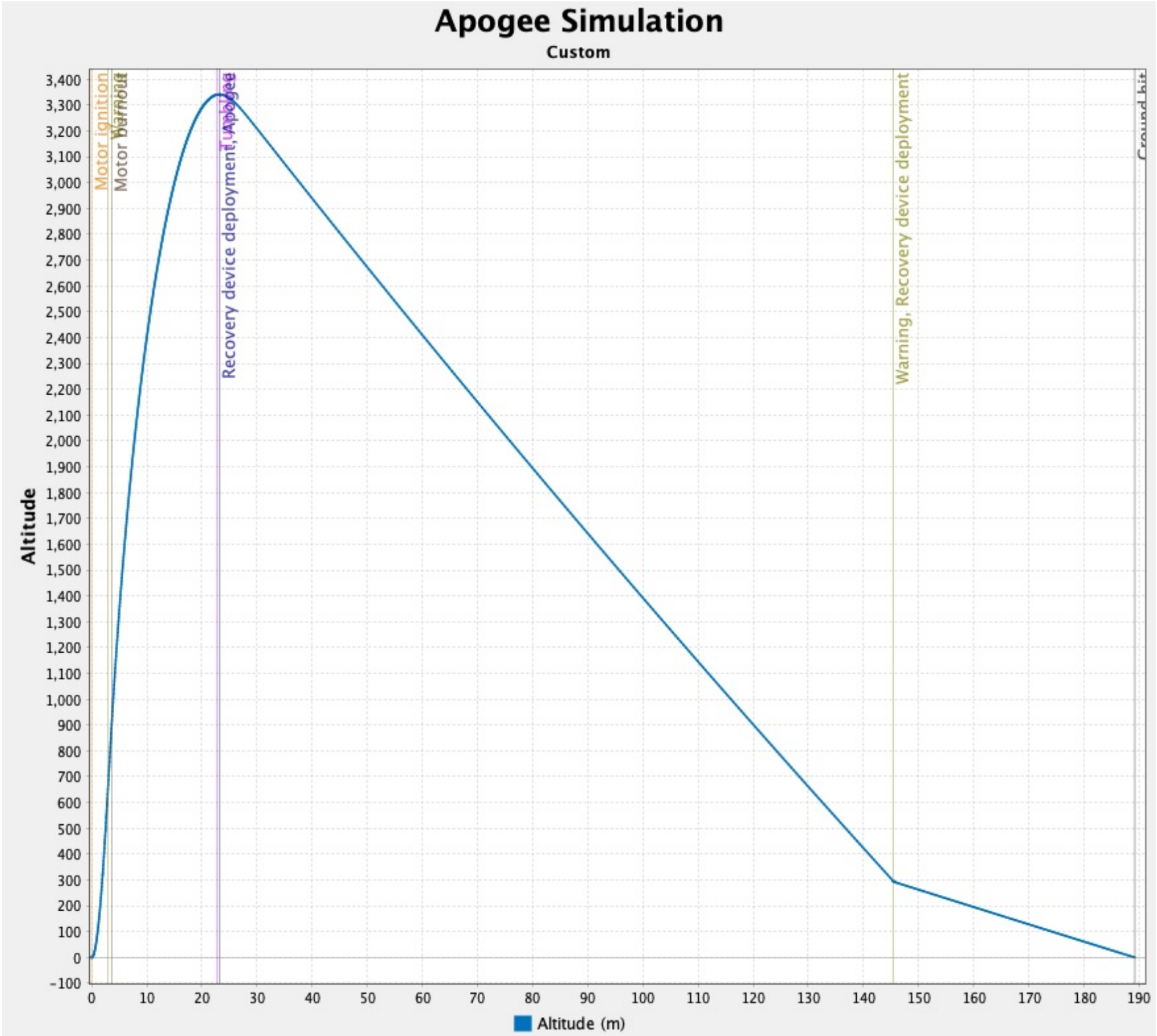
Model Notes

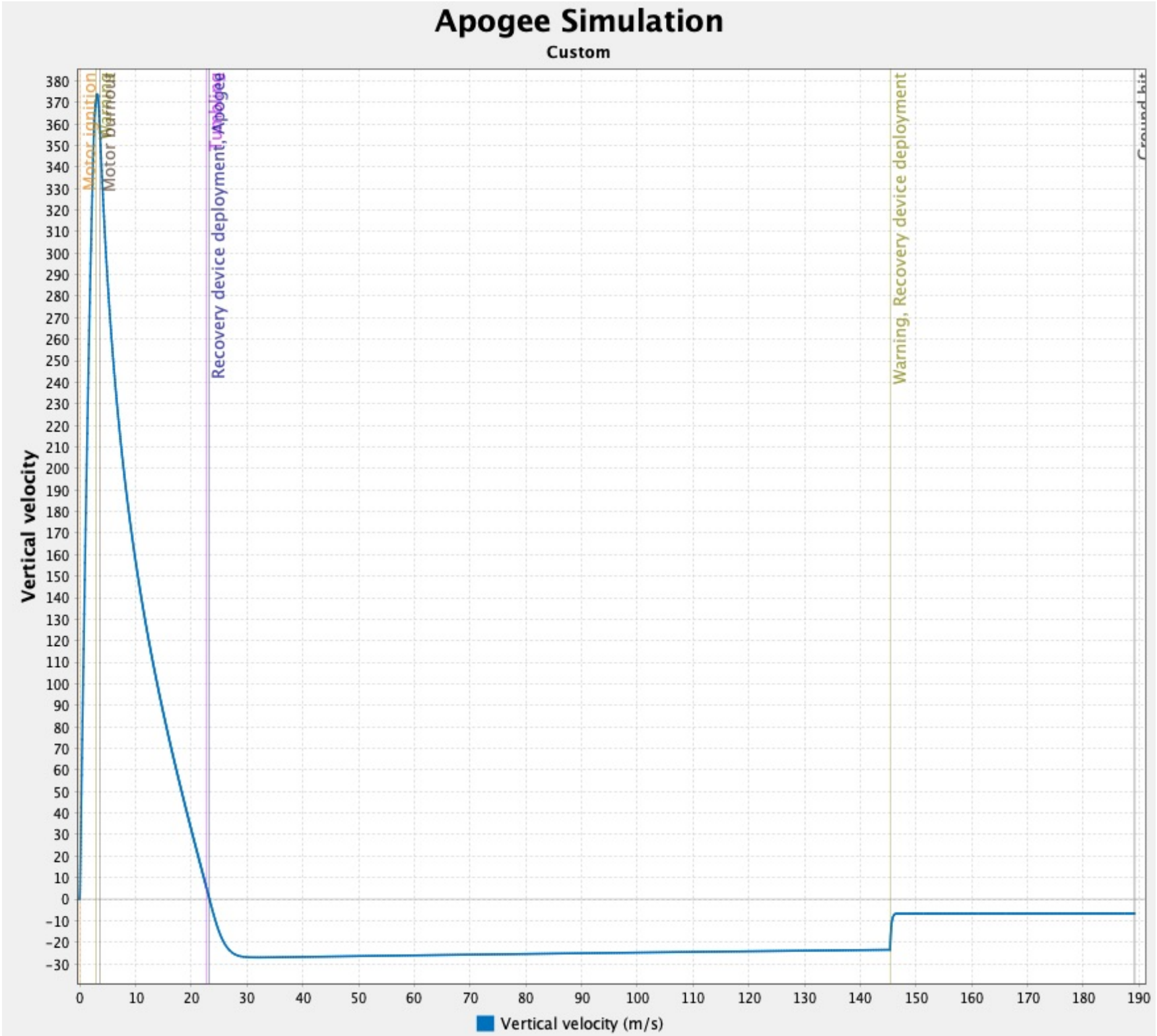
- Ascent only; no parachute descent
- Vertical flight, zero wind, constant drag coefficient
- Density decreases with altitude using troposphere model
- Motor and simulation data from OpenRocket

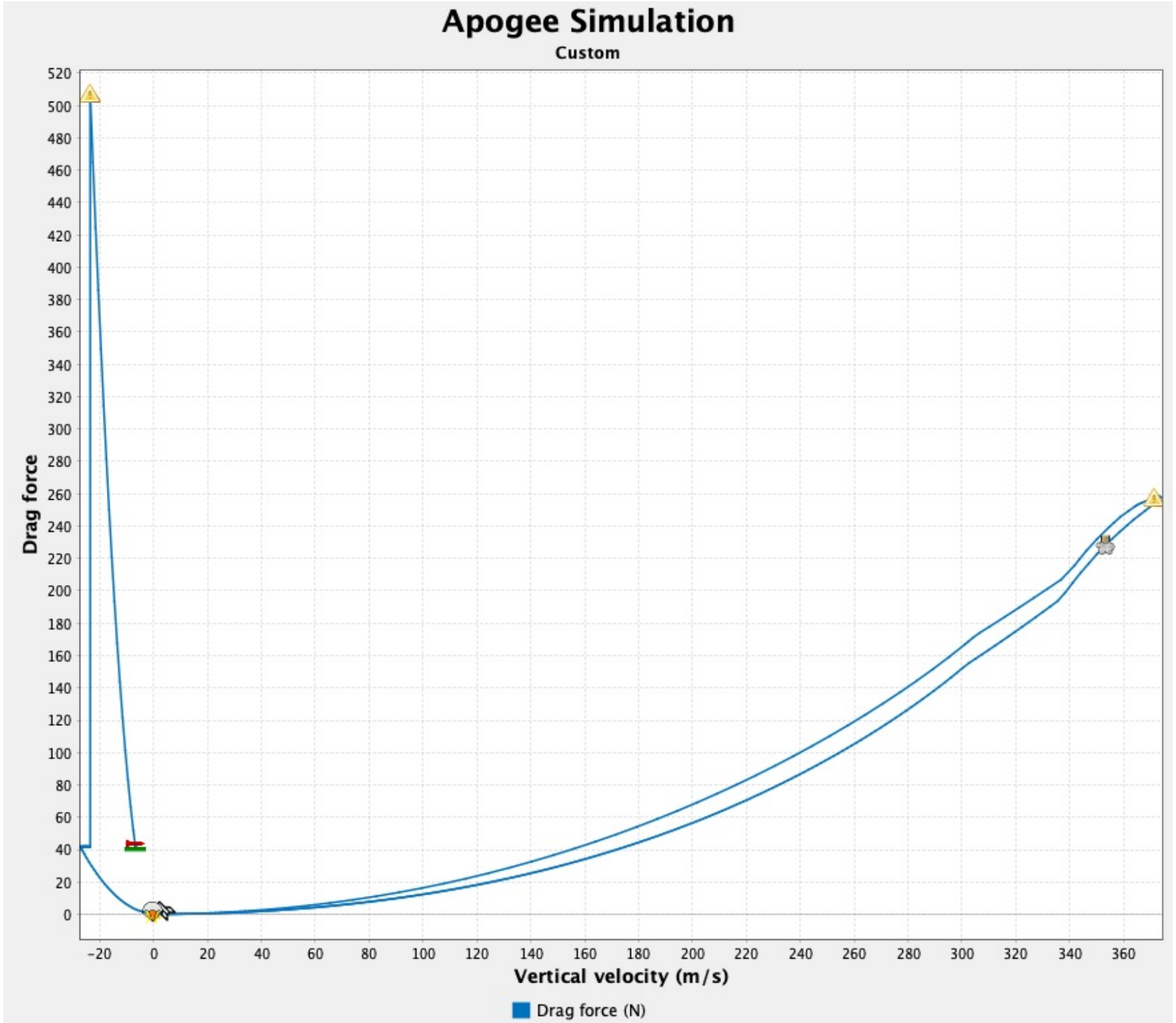
Comparison

The Euler's method calculator predicted a slightly higher apogee of 3589 m in comparison to the OpenRocket simulation apogee of 3342 m (approximate difference of 7.4%). The calculator also predicted a higher max q of 90.6 kPa in comparison to the simulated maximum dynamic pressure of 80.0 kPa (approximate difference of 13.2%).

There are a few simplifications and assumptions used in the calculator that can be used to explain these differences. First, excel uses a constant drag coefficient, while OpenRocket allows for the coefficient change throughout the simulation. Second, excel assumes the rocket flies straight up with no wind, an unrealistic simplification for real world applications. Since we know that the drag coefficient actually changes in real time along with the constantly changing flight conditions, and real-world conditions such as wind and changes in flight direction exist and should be accounted for to obtain accurate simulations, OpenRocket provides us with the more accurate predictions thus accounting for some of the difference. Lastly, the Euler calculator uses simple step sizes, which introduce some numerical error. OpenRocket however, uses a more accurate calculation method, helping to further explain for the differing predictions.


Apogee Simulation
Custom
Altitude
Motor ignition
Motor burnout
Recovery device deployment, Apogee
Warning, Recovery device deployment
Altitude (m)


Apogee Simulation
Custom
Vertical velocity
Motor ignition
Motor burnout
Recovery device deployment, Apogee
Warning, Recovery device deployment
Vertical velocity (m/s)


Apogee Simulation
Custom
Drag force
520
500
480
460
440
420
400
380
360
340
320
300
280
260
240
220
200
180
160
140
120
100
80
60
40
20
0
-20
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
320
340
360
Vertical velocity (m/s)
Drag force (N)